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Calibrating Reliance on Automated Advice: Transparency and Trust Calibration Feedback

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ABSTRACT

Inappropriate reliance on automated advice can result in humans accepting incorrect or rejecting correct advice. Increased automation transparency and trust calibration feedback are principles purported to promote accurate automation use. We examined the effects of automation transparency, trust calibration feedback, and their potential interacting effect on automation use accuracy and other outcomes. Participants completed uninhabited vehicle management missions by agreeing/disagreeing with automated advice. Transparency was manipulated within-subjects (low, high) and trust calibration feedback between-subjects (absent, present). If trust was inappropriate, trust calibration feedback instructed participants to take their time and carefully check display information. Higher transparency benefited automation use accuracy, decision time, perceived workload, trust, and usability. Trust calibration feedback had no benefit on automation use accuracy and subsequently did not amplify the benefits of increased transparency. These findings have potential implications to inform the design of automated decision aids to support human understanding of, and calibration to, automation capabilities.

KEYWORDS

Automation and human cognition; automation transparency; trust feedback; uninhabited vehicle control; decision-support systems

1. Introduction

Automated decision aids are commonly used in work settings such as the military, aviation, healthcare, and cybersecurity (National Academies of Sciences, Engineering, & Medicine (NASEM), 2022). As automation becomes more complex, humans can find it difficult to understand and verify automated advice, posing challenges to estimating the automation's reliability and appropriately calibrating trust in automation. Numerous factors such as human characteristics (e.g., personality), situational elements (e.g., task difficulty), and automation design (e.g., predictability) have been shown to impact trust in and reliance on automation (Hancock et al., 2011). Although evidence suggests there is not an one-to-one mapping between trust in automation and reliance on automation (Patton & Wickens, 2024), under-trust increases the probability of under-relying on automation and *disusing* automated advice (rejecting correct advice), whereas excessive trust increases the probability of over-reliance and *misusing* automated advice (accepting incorrect advice; Lee & See, 2004; Parasuraman & Riley, 1997). Mosier and Skitka (1996) proposed that humans can incorrectly accept incorrect advice (misuse) due to *automation bias* – defined as humans using the outcome of the decision aid "as a heuristic replacement for vigilant information seeking and processing" (p. 205). Therefore, a key challenge concerns fostering calibrated trust in the automation, that is, ensuring humans align their trust with the automation's true capabilities/reliability (Carter et al., 2024; Strickland et al., 2024; Wischnewski et al., 2023), such that disuse and misuse can be minimized.

Providing continuous information to aid human understanding of the automation and its capabilities has been emphasized as a work design principle to assist trust calibration and mitigate over/under-reliance on advice (Wischnewski et al., 2023). Increased *transparency* information about the reasoning behind automated advice is one such method that has been found to increase the accuracy of automation use (see reviews by Bhaskara et al., 2020; Sargent et al., 2023; Van de Merwe, Mallam, & Nazir, 2024; and for examples of related work in the explainable artificial intelligence (AI) literature see Naiseh et al., 2023; Zhang et al., 2020). Providing *trust calibration* feedback (e.g., human/automation performance-oriented feedback) has also been suggested as a work design principle to promote trust calibration and accurate automation use (e.g., McGuirl & Sarter, 2006; Okamura & Yamada, 2020). Both methods are intended to encourage humans to critically evaluate the trust they are placing in automated advice. Together then, transparency and trust calibration feedback may interact to further improve the accuracy of automation use. Trust calibration feedback may direct human attention towards increased transparency information when most necessary, to

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better assist with understanding the reasoning behind advice and whether to accept/reject it. The current study examined the effects of transparency and trust calibration feedback on the accuracy of automation use and other outcome variables.

1.1. Automation transparency

Automation transparency has been defined as providing "a real-time understanding of the actions of the AI system" (National Academies of Sciences, Engineering, & Medicine (NASEM), 2022, p. 31), in order to increase the "understandability and predictability of [a] system" (Endsley et al., 2003, p. 146). In the explainable AI literature this concept is typically referred to as providing "local explanations" for automated/AI advice (e.g., Dwivedi et al., 2023; Naiseh et al., 2023). The Situation-Awareness Agent-Based Transparency (SAT) model proposes three levels of transparency: Level 1 – purpose and intentions, Level 2 – rationale behind advice, and Level 3 – projected outcomes if advice is followed (Chen et al., 2014). Despite the prominence of the SAT model for designing transparent automation, the specific task domain where automation is applied impacts how transparency is operationalized and designed (Skraaning & Jamieson, 2021; Van de Merwe, Mallam, & Nazir, 2024; Van de Merwe, Mallam, Nazir, & Engelhardtsen, 2024). Typically, increased transparency leads to more accurate automation use, without costs, or even with improvements to decision time and perceived workload and can foster higher perceived trust and usability (for reviews see Bhaskara et al., 2020; Sargent et al., 2023; Van de Merwe, Mallam, & Nazir, 2024). These findings have also been found in uninhabited vehicle (UV) management task settings (Gegoff et al., 2024, 2025; Mercado et al., 2016; Stowers et al., 2020; Tatasciore et al., 2023, 2024; Tatasciore & Loft, 2024), which is the task used here due to its high relevance to military settings.

While the benefits of increased transparency are well documented, several studies have found a bias towards agreeing with automated advice, which can result in no improvements to rejecting incorrect advice (Tatasciore et al., 2023, 2024; Tatasciore & Loft, 2024), or even poorer correct rejection rates and poorer overall accuracy of automation use (Bhaskara et al., 2021; Bussone et al., 2015). This indicates that in these prior studies individuals may not have always thoroughly examined the provided high transparency information to verify automated advice. This interpretation is in line with prior research reporting automation bias by professional pilots and novice participants (Mosier et al., 1992; Skitka et al., 1999; for related research see Layton et al., 1994; Molloy & Parasuraman, 1996). These studies indicate that humans prefer to use minimal cognitive effort in decision making (cognitive-miser hypothesis; Wickens & Hollands, 2000), often accepting and actioning automated advice despite contra-indications from other sources of information of which they are aware. The provision of increased transparency may amplify automation bias if participants ascribe greater analysis capability and power/authority to automation that provides an increased rationale (local explanation) for its recommendations (Dwivedi et al., 2023; Lee & See, 2004).

Presenting trust calibration feedback alongside increased transparency may assist operators with better understanding the automation's capabilities, and therefore minimize automation bias and inappropriate reliance that can occur with higher transparency provision. Specifically, trust calibration feedback could potentially amplify the benefits of increased transparency on the accuracy of automation use.

1.2. Trust calibrations

Prior studies have examined the effects of presenting different forms of human and/or automation performance-oriented feedback. Automation performance-oriented trust calibration feedback is intended to inform humans about the capabilities and limitations of a system, rather than the automation's inner workings. For example, McGuirl and Sarter (2006) found that presenting the automation's confidence in its advice on a case-by-case basis improved trust calibration, reduced automation bias, and increased the accuracy of airline pilot decision-making. Relatedly, Seong and Bisantz (2008) found that providing information regarding the validity of automated estimates/advice (i.e., the consistency of estimates/advice) improved trust calibration, as indicated by more accurate automation use. However, contrary to such findings, Tatasciore et al. (2024) reported that case-by-case automation confidence information caused a symmetrical (uncalibrated) shift in response bias with individuals more likely to agree with more highly confident automation, resulting in no net benefit to the accuracy of automation use. Similarly, Zhang et al. (2020) found no benefit of providing automation confidence on the accuracy of AI advice use.

Several studies have examined the impact of human performance-oriented feedback following an interaction (i.e., experience with automation) on trust calibration. However, findings are mixed. For example, Lu and Sarter (2019) found no impact of human or automation performance-oriented feedback (i.e., positive/negative sound tone as a function of operator/automation accuracy) on perceived trust or performance. However, Chen et al. (2021) found that human performance-oriented feedback (i.e., human decision accuracy) improved automation use (decision sensitivity), led to faster decision times, and alleviated the bias toward agreeing with automated advice (also see Chen et al., 2018).

Trust calibration feedback which highlights whether reliance on automated advice is appropriate (or not) has been proposed as another way to mitigate inappropriate trust (Okamura & Yamada, 2020). Such feedback is intended to alert and prompt the human to recognize inaccurate use of automation, and to adjust their trust and automation reliance in the future. Okamura and Yamada (2020) examined the effect of trust calibration feedback cues (i.e., visual, audio, verbal, or anthropomorphic) presented when the automation detected inappropriate trust in automation via human choices. Inappropriate trust was detected when the

human accepted incorrect automated advice. They found that presenting trust calibration feedback cues when inappropriate trust was detected, as opposed to the presentation of real-time cumulative automation reliability information, led to improved automation use.

The current study further explores the effects of human and automation performance-oriented trust calibration feedback on the accuracy of automation use. Specifically, we were interested in the extent to which trust calibration feedback presented after each interaction with automation (i.e., trial) could improve the accuracy of automation use. In contrast to most prior studies (e.g., Chen et al., 2021; Lu & Sarter, 2019), trust calibration feedback in the current study provided information about both human and automation performance (Okamura & Yamada, 2020). It is worth noting that in our prior UV management task studies (e.g., Gegoff et al., 2024, 2025; Tatasciore et al., 2023; 2024; Tatasciore & Loft, 2024), trial-by-trial feedback was provided regarding human (i.e., decision) and automation accuracy (i.e., how reliable the automation was on the previous trial), and thus it is appropriate that this type of feedback formed the baseline comparison condition for the current study. However, in the current study, trust calibration feedback also explicitly specified whether human trust in automation was appropriate or not on a trial-by-trial basis. Critically, when trust was inappropriate, the feedback also explicitly instructed individuals to take their time and check the information on the display more carefully before making future decisions.

We predicted that this form of trust calibration feedback should encourage humans to re-evaluate their current calibration to the automation's capabilities, and thus should increase the accuracy of automation use. It is possible, however, that the provision of trust calibration feedback may also increase correct decision time or perceived workload. It has been consistently found in the decision-making literature that decision times are slower on trials following an incorrect response compared to trials following a correct response, a phenomenon referred to as "post-error slowing" (e.g., Fievez et al., 2022; Fu et al., 2019). In line with this, we predicted that decision times would be slower on the trial following incorrect compared to correct automation use, but that this effect may be more pronounced when provided trust calibration feedback.

1.3. Automation transparency and trust calibrations

Transparency and trust calibration feedback may interact to further benefit the accuracy of automation use. Increased transparency can increase the accuracy of automation use (hit rates, sensitivity), but does not often improve correct rejection rates due to the bias towards agreeing with automated advice (e.g., Tatasciore et al., 2023; Tatasciore & Loft, 2024). Trust calibration feedback may reduce the bias associated with increased transparency provision, given that it should assist users in understanding and re-evaluating their degree of calibration to the automation's capabilities.

The current study examined potential interacting effects between transparency and trust calibration feedback in a UV management task that required participants to deploy the optimal UV to complete missions by accepting/rejecting automated advice. To our knowledge, no prior studies have examined this. However, a recent study by Tatasciore et al. (2024) using a similar UV management task examined the potential interaction between the provision of transparency and case-by-case automation confidence information. Tatasciore et al. found a symmetric (uncalibrated) shift in bias from agreeing with highly confident advice to disagreeing with less confident advice, resulting in no net benefit of automation confidence on the accuracy of automation use. This uncalibrated shift in bias was not mitigated by increased transparency and thus automation confidence information did not interact with transparency to further improve automation use. In the current study, when trust was inappropriate, trust calibration feedback instructed participants to take their time to scrutinize transparency information before making future decisions. Thus, when trust calibration feedback is provided, higher transparency may lead to even further improved accuracy of automation use compared to lower transparency, because users will be prompted to more carefully assess the presumably more useful highly transparent information when most necessary (i.e., when their trust in automation has been historically inappropriate).

2. Methods

2.1. Participants

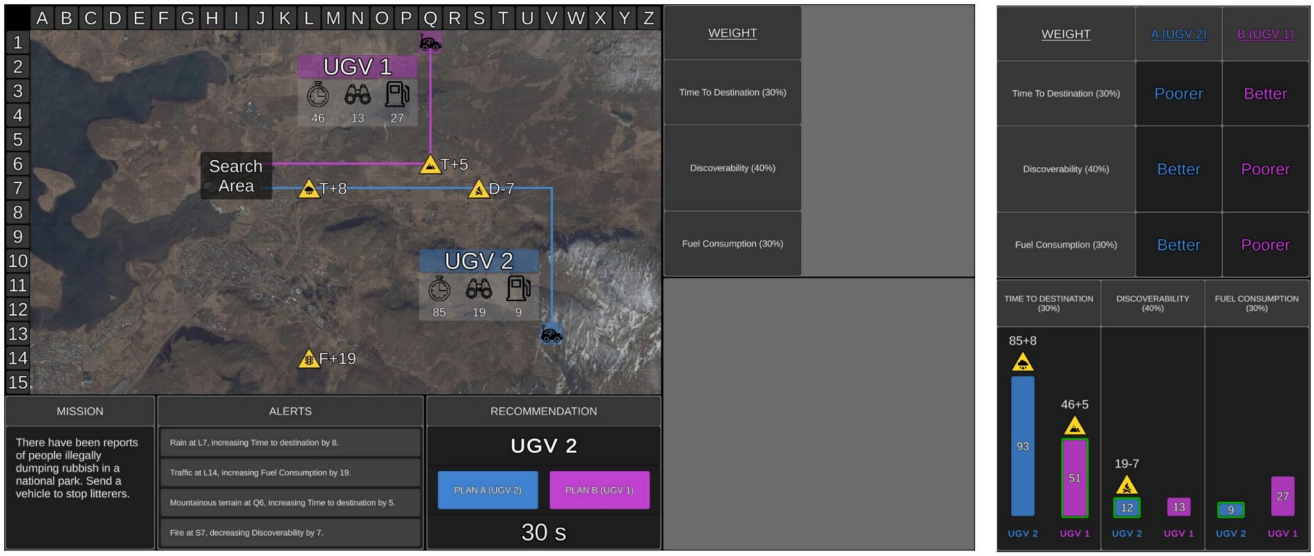
Participants were 160 undergraduate students (98 females, 62 males; $M = 19.18$) who participated in exchange for course credit and were given a performance incentive of up to AUD$30. Participants were randomly assigned to either the trust calibration feedback present ($N = 80$) or absent ($N = 80$) conditions.

2.2. Design

A mixed design was used with transparency (low, high) as the within-subjects factor, and trust calibration feedback (present, absent) as the between-subjects factor.

2.3. Uninhabited vehicle management task

The UV management task and transparency designs were informed via consultation with the Australian Defense Science and Technology Group (i.e., observation and discussion of UV management platforms). The UV management task was displayed on a single desktop monitor (Figure 1). A total of 200 missions (trials) were split into two blocks of 100 trials, with one block completed with low transparency and the other with high transparency. Mission statements were provided but were not relevant to optimal UV selection. A tactical map of either rural, coastal, or urban areas was presented. On the tactical map was the mission search area, two UVs (aerial [UAV]; ground [UGV]; or surface [USV]) which were numbered 1 or 2 (no particular order),



(a) (b)

Figure 1. An example mission with (a) low transparency and (b) high transparency. *Note:* On the tactical map are two UV's and their capability scores, environmental factor symbols, and the path each UV will take to reach the search area. The Recommender has correctly advised UGV 2 as Plan A. Transparency information is presented in the table (top right) and graph (bottom right; high transparency only) displays. Table rows are larger for higher weighted capabilities, signifying how the Recommender evaluated the importance of each capability. In the table, high transparency (b) also presents which UV the Recommender considered to be better and poorer on each capability. In the graph, high transparency illustrates the Recommender's final calculated score including which factors it considered and their level of impact, with the bar of better UV capabilities outlined in green. The tactical map, mission, alerts, and recommendation windows are identical regardless of transparency condition. Adapted from Tatasciore et al. (2023).

and a line illustrating the path each UV would take to reach the search area.

To select the optimal UV, participants considered UV capabilities, capability importance weightings, and environmental factors impacting capabilities. Capabilities included time to destination (UV duration to reach the search area), discoverability (how discoverable the UV was by third parties), and fuel consumption (fuel consumed to reach the search area). Capabilities were presented as values displayed adjacent to either a timer (time to destination), binocular (discoverability), or fuel gauge (fuel consumption) symbol on the tactical display (Figure 1). A lower score on each capability was better (i.e., quicker, less discoverable, and less fuel consumed).

One of five importance weighting sets was displayed in the weightings window during each mission (Figure 1). Hard sets included two capabilities with equal weightings (e.g., 45, 45, 10%), where participants considered which UV scored lower on two capabilities. Easy sets included one highly weighted capability (e.g., 60, 20, 20%), where participants considered which UV scored lower on the highly weighted capability.

Four environmental factors were presented during each mission as yellow environmental factor symbols, which were relevant (i.e., presented on the path of an UV) or irrelevant (i.e., not presented on a UV path and could be ignored) to UV capabilities (Figure 1). For each mission, there were one to three relevant factors. Factor symbols showed the capability impacted (T = time to destination, D = discoverability, F = fuel consumption), type of factor (e.g., fog), direction of impact (+ =positive, − =negative), and the impact value. The alerts window also conveyed messages regarding factors and their impact.

In the recommendation window (Figure 1), the Recommender advised the optimal UV as Plan A, after considering capabilities, their weightings, and relevant environmental factors, and the other UV as Plan B. Plan A was positioned on the left-hand side (and coloured blue), and Plan B on the right-hand side (purple). Participants were required to agree with the Recommender and select Plan A or to instead select Plan B within 30s (maximum). If the participant did not respond within this time, an incorrect response was recorded, and the next trial began.

The 200 trials were split into two blocks of 100 trials. Both blocks included a similar number of capability weighting sets, relevant environmental factors, factor impacts, and error types (see below). Block presentation order was counterbalanced, as well as whether low/high transparency was presented first. The order of unique trials was randomized and yoked such that sets of four participants received the same randomized order of trials, with only two of the four participants receiving trust feedback.

2.3.1. Transparency manipulation

Low transparency information (table display; Figure 1(a)), broadly equating to SAT Level 1 (automation's goals/intent), displayed how the Recommender assessed the importance of UV capabilities, with weighting rows larger for higher weighted capabilities.

With high transparency information (broadly equating to SAT Level 1 + 2 + 3; rationale and projected outcomes), the table display also indicated which UV the Recommender considered to be "better" and "poorer" on each capability (Figure 1(b)). A graph display also presented *how* the Recommender calculated capability scores for each UV and

the projected level of impact of the environmental factors it considered in its calculation. Specifically, when the Recommender considered a relevant factor, that factor symbol was placed on top of the relevant bar on the graph. Additionally, the factor impact value that was added or subtracted from the original capability score was presented above the factor symbol. The Recommender's calculated score was also presented within the respective bar. Shorter bars demonstrated better capability as lower scores were considered better. For each capability, a green border was placed around the bar of the UV that the Recommender had considered would perform better on that capability. High transparency therefore presented information about how the Recommender made its calculations (SAT Level 2) and the Recommender's projection of the consequences of variability in the task environment and projected outcomes if its advice was actioned (SAT Level 3).

The Recommender was 75% reliable, advising the incorrect UV as Plan A on 25% of the trials. Participants were notified that "The recommender is only reasonably reliable automation, and as such it may not always recommend the optimal plan." When the advice was incorrect, the Recommender either: missed a relevant environmental factor, miscalculated a relevant factor impact, or a combination of these errors across different relevant factors.

If the Recommender missed a relevant environmental factor, the factor symbol would be missing from the graph display (high transparency) and the original capability score would be incorrectly presented within the relevant bar. When the Recommender miscalculated a relevant factor impact, the factor symbol would be presented in the graph display (high transparency), but the score that the Recommender added/subtracted would be incorrect, resulting in an incorrect final score. As a result of these errors, the green border may have been placed around the incorrect UVs bar on the graph, and the UV that the Recommender considered to be "better" and "poorer" in the table display may have also been incorrect.

On 10% of the reliable advice trials, the Recommender advised the correct UV, but still made an error/s in its calculation, but these errors were not substantial enough to cause the Recommender to advise the incorrect UV. These trials were included to avoid participants simply selecting Plan B as soon as they noticed any type of error (e.g., Gegoff et al., 2024).

2.3.2. Trust calibration manipulation

When trust calibration feedback was absent, feedback after each mission only regarded human and automation performance accuracy (Figure 2), as in our prior studies using the UV management task (e.g., Gegoff et al., 2024; Tatasciore et al., 2023; Tatasciore & Loft, 2024). This performance feedback provided indirect feedback on participant trust, but the trust calibration feedback (detailed below) made the importance of calibrated automation trust more salient and provided instructions for participant actions in future trials.

When trust calibration feedback was presented (Figure 3), participants also received feedback regarding their trust in the Recommender following each mission. When trust was appropriate (Figure 3(a,b)), the message "trust appropriate – correctly agreed/disagreed with the Recommender" was presented in green, and when trust was inappropriate, the message "trust inappropriate – incorrectly agreed/disagreed with the Recommender" was presented in red (Figure 3(c,d)). In addition, when trust was inappropriate, with low transparency participants were instructed to "Please take your time and check the information on the display more carefully before making decisions in the future," and with high transparency to "Please take your time and check the high transparency information more carefully before making decisions in the future."

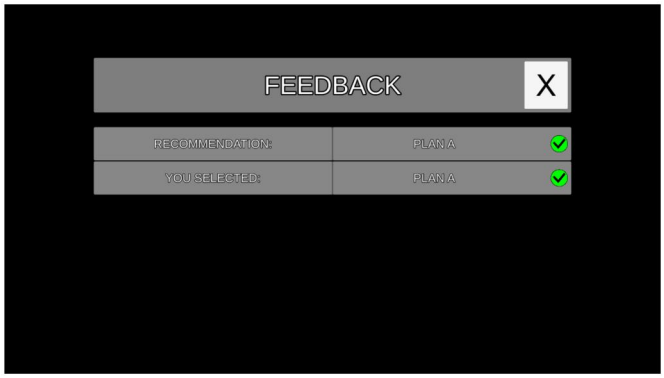


Figure 2. Example feedback screen when trust calibration feedback was absent. *Note:* In this example, the automated advice (recommendation) and human decision were correct and represented by the green ticks. When the automated advice or human decision were incorrect a red cross was instead presented. Feedback presentation was identical regardless of whether low or high transparency was provided.

2.4. Measures

2.4.1. Automated task performance

Hit rate was the proportion of trials that Plan A was correctly selected, whilst correct rejection rate was the proportion of trials that Plan B was correctly selected. The Signal Detection metric d' measured sensitivity to correctly discriminate Plan A from B. The criterion (c) metric measured bias towards agreeing with the automated advice (negative values = higher bias to agree with automation). We adjusted for extreme hit or false alarm values (0 or 1), with rates of 0 replaced with 0.5/n, and rates of 1 replaced with (n − 0.5)/n, where n is the number of signal (Plan A correct) or noise (Plan B correct) trials (Macmillan & Kaplan, 1985). Decision times were based on correct decisions.

2.4.2. Workload

Subjective workload was measured using the Air Traffic Workload Input Technique (ATWIT; Stein, 1985), with participants rating their workload from 1 (very low) to 10 (very high) after each mission.

2.4.3. Trust

Perceived trust was measured using a six-item questionnaire adapted from Merritt (2011). Items were rated on a 5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree).

(a) Trust appropriate, correctly agreed

(b) Trust appropriate, correctly disagreed

(c) Trust inappropriate, incorrectly agreed

(d) Trust inappropriate, incorrectly disagreed

Figure 3. Example feedback screens when trust calibration feedback was present with high transparency. (a) Trust appropriate, correctly agreed. (b) Trust appropriate, correctly disagreed. (c) Trust inappropriate, incorrectly agreed. (d) Trust inappropriate, incorrectly disagreed. *Note:* The table at the bottom of each screen presents human and automation performance (accuracy) information with green ticks representing correct advice/human decision and red crosses representing incorrect advice/human decision. When the human decision was correct (a and b), the trust appropriate message was presented in green, whereas when the human decision was incorrect (c and d), the trust inappropriate message was presented in red. These messages were identical for the low and high transparency conditions. However, the message in white varied slightly for the low transparency condition with participants being told to instead check the "information on the display."

2.4.4. System Usability Scale

Usability was measured using an adapted version of the 10-item System Usability Scale (SUS; Brooke, 1996). Items were rated on a 5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree). Some items were reverse-scored and then all items were added and multiplied by 2.5 to yield a score from 0 to 100.

2.4.5. Trust Calibration Feedback Questionnaire

To explore how participants used the trust calibration feedback, a two-item open-ended questionnaire was used. Items were "How did you use the trust feedback?" and "If your trust was inappropriate, what steps did you take after receiving this feedback?"

2.5. Procedure

Participants first viewed a 20 min audio-visual PowerPoint presentation explaining how to complete the task manually, followed by 20 manual practice trials. Participants then watched a PowerPoint specific to the transparency assigned during Block 1. If trust calibration feedback was provided, information regarding how to interpret each of the possible types of appropriate and inappropriate feedback was included in this presentation. Participants then completed 100-test trials with a 60s (minimum) break after 50 trials.

Table 1. Manual training descriptive statistics as a function of trust calibration feedback condition.

	Trust calibration feedback	
	Absent	Present
UV selection accuracy	0.85 (0.14)	0.86 (0.10)
Decision time (s)	15.67 (3.64)	15.59 (3.00)
Perceived workload	4.05 (1.41)	3.88 (1.28)

Note: Standard deviations are presented in parentheses. s: seconds.

Participants then completed the trust and SUS questionnaires (counterbalanced order). Before Block 2 trials participants watched another PowerPoint explaining the transparency available in Block 2. They then completed Block 2 (60s break after 50 trials), followed by the trust and SUS questionnaires. Lastly, they completed the trust calibration feedback questionnaire. The total experiment duration was 2 hrs.

3. Results

Data were removed for two participants who did not follow task instructions (one in each of the trust calibration feedback conditions). Manual training descriptive statistics as a function of trust calibration feedback condition are presented in Table 1, with no significant differences on outcome variables (smallest $p = 0.43$).

Descriptive statistics for outcome variables during test trials, as a function of transparency and trust calibration

Table 2. Test trial descriptive statistics as a function of transparency and trust calibration feedback.

	Hit	CR	d'	c	DT	Workload	Trust	Usability
Trust calibration feedback absent								
Low transparency	0.88 (0.10)	0.81 (0.18)	20.24 (10.07)	−0.13 (0.21)	12.60s (3.30s)	4.25 (1.48)	2.42 (0.77)	61.67 (14.68)
High transparency	0.93 (0.09)	0.83 (0.17)	20.70 (10.04)	−0.30 (0.28)	11.46s (3.65s)	3.72 (1.33)	2.59 (0.80)	65.00 (15.48)
Trust calibration feedback present								
Low transparency	0.87 (0.09)	0.79 (0.17)	2.13 (1.00)	−0.17 (0.19)	13.19s (3.17s)	4.46 (1.53)	2.46 (0.80)	58.70 (14.79)
High transparency	0.93 (0.09)	0.80 (0.20)	2.60 (1.13)	−0.35 (0.33)	12.02s (3.70s)	4.03 (1.51)	2.75 (0.85)	61.71 (15.21)

Note: Standard deviations are presented in parentheses. CR: correct rejection rates; DT: decision time; d': sensitivity; c: criterion; s: seconds.

feedback, are presented in Table 2. A series of 2 Transparency (low, high) x 2 Trust Calibration Feedback (absent, present) mixed ANOVAs were run, with plans to follow-up interactions between transparency and trust calibration feedback with t-tests comparing low to high transparency when trust calibration feedback was absent and present. Partial eta squared (small = 0.01, medium = 0.06, large = 0.14) was used to estimate effect sizes for F-tests (Cohen, 1992).

3.1. Trust calibration feedback rates

For the trust calibration feedback present condition, inappropriate trust calibration feedback was presented on 15% ($SD = 0.10$) of the trials with low transparency and 12% ($SD = 0.14$) of the trials with high transparency, reflecting inaccurate automation use proportions for the two within-subject transparency conditions.

3.2. Automation use

3.2.1. Hit rates

There was a main effect of transparency, $F(1,156) = 56.35$ $p < 0.001$, $\eta_\rho^2 = 0.27$, with higher hit rate with high ($M = 0.93$, $SD = 0.09$) compared to low ($M = 0.88$, $SD = 0.09$) transparency. There was no main effect of trust calibration feedback, $F < 1$, and no interaction, $F < 1$.

3.2.2. Correct rejection rates

There was no main effect of transparency, $F(1,156) = 1.76$, $p = 0.19$, or trust calibration feedback, $F < 1$, and no interaction, $F < 1$.

3.2.3. Sensitivity

There was a main effect of transparency, $F(1,156) = 41.85$, $p < 0.001$, $\eta_\rho^2 = 0.21$, with higher sensitivity with high ($M = 2.65$, $SD = 1.08$) compared to low ($M = 2.18$, $SD = 1.03$) transparency. There was no main effect of trust calibration feedback, $F < 1$, and no interaction, $F < 1$.

3.2.4. Criterion

There was a main effect of transparency, $F(1,156) = 44.18$, $p < 0.001$, $\eta_\rho^2 = 0.22$, with a greater bias towards agreeing with automation with high ($M = -0.33$, $SD = 0.31$) compared to low ($M = -0.15$, $SD = 0.20$) transparency. There was no main effect of trust calibration feedback, $F(1,156) = 1.61$, $p = 0.21$, and no interaction, $F < 1$.

3.3. Correct decision time

There was a main effect of transparency, $F(1,156) = 14.36$, $p < 0.001$, $\eta_\rho^2 = 0.08$, with quicker correct decisions with high ($M = 11.74$s, $SD = 3.67$s) compared to low ($M = 12.90$s, $SD = 3.24$s) transparency. There was no main effect of trust calibration feedback, $F(1,156) = 1.57$, $p = 0.21$, and no interaction, $F < 1$.

3.4. Perceived workload

There was a main effect of transparency, $F(1,156) = 41.53$, $p < .001$, $\eta_\rho^2 = 0.21$, with lower perceived workload with high ($M = 3.88$, $SD = 1.43$) compared to low ($M = 4.35$, $SD = 1.50$) transparency. There was no main effect of trust calibration feedback, $F(1,156) = 1.41$, $p = 0.24$, and no interaction, $F < 1$.

3.5. Trust and usability

For perceived trust, there was a main effect of transparency, $F(1,156) = 15.13$, $p < 0.001$, $\eta_\rho^2 = 0.09$, with higher trust ratings with high ($M = 2.67$, $SD = 0.83$) compared to low ($M = 2.44$, $SD = 0.78$) transparency. There was no main effect of trust calibration feedback, $F(1,156) = 1.10$, $p = 0.30$, and no interaction, $F(1,156) = 1.10$, $p = 0.30$.

For perceived usability, there was a main effect of transparency, $F(1,155) = 5.82$, $p = 0.02$, $\eta_\rho^2 = 0.04$, with higher usability ratings with high ($M = 63.34$, $SD = 15.38$) compared to low ($M = 60.18$, $SD = 14.76$) transparency. There was no main effect of trust calibration feedback, $F < 1$, and no interaction, $F < 1$.

3.6. Decision time following incorrect compared to correct automation use

To examine whether there was a difference in decision times on the trial following incorrect compared to correct automation use we conducted a 2 Transparency (low, high) × 2 Trust Calibration Feedback (absent, present) × 2 Automation Use (correct, incorrect) mixed ANOVA. Trust calibration feedback was the between-subjects factor, whilst transparency and automation use were within-subject factors. Table 3 presents the descriptive statistics.

There was a main effect of transparency, $F(1,152) = 10.51$, $p = 0.001$, $\eta_\rho^2 = 0.07$, with faster decisions with high ($M = 12.98$s, $SD = 4.08$s) compared to low ($M = 14.11$s, $SD = 3.53$s) transparency. There was also a main effect of automation use, $F(1,152) = 152.16$, $p < 0.001$,

Table 3. Mean decision times on the trial following correct and incorrect automation use as a function of transparency and trust calibration feedback.

	Correct automation use	Incorrect automation use
Trust calibration feedback absent		
Low transparency	12.66s (3.30s)	14.87s (4.16s)
High transparency	11.54s (3.69s)	13.57s (4.48s)
Trust calibration feedback present		
Low transparency	13.21s (3.33s)	15.56s (4.28s)
High transparency	12.06s (3.70s)	14.64s (5.34s)

Note: Standard deviations are presented in parentheses. s: seconds. The last trial from each block was excluded from these analyses as there was no subsequent trial.

$\eta_\rho^2 = 0.50$, with slower decisions on the trial following incorrect ($M = 14.63$s, $SD = 3.78$s) compared to correct automation use ($M = 12.37$s, $SD = 2.91$s). There was no main effect of trust calibration feedback, $F(1,152) = 1.76$, $p = 0.19$. There were also no interactions between transparency and trust calibration feedback, transparency and automaton use, automation use and trust calibration feedback, and no three-way interaction, all Fs < 1.

4. General discussion

The current study examined the effects of automation transparency (low, high) and trust calibration feedback on the accuracy of automation use and other outcome variables. We predicted that transparency and trust calibration feedback would interact to further benefit the accuracy of automation use.

4.1. Automation transparency

Increased transparency led to more accurate automation use in terms of hit rate and sensitivity, quicker correct decisions, lower perceived workload, and higher perceived trust and usability. However, there were no benefits to correct rejection rates, and there was a bias towards agreeing with *advice* with increased transparency. The finding however that, despite the observed automation bias, increased transparency benefitted the accuracy of automation use replicates prior UV management studies (e.g., Gegoff et al., 2024, 2025; Mercado et al., 2016; Stowers et al., 2020; Tatasciore et al., 2023; 2024; Tatasciore & Loft, 2024), as well as prior findings across other task domains (e.g., air traffic control, process control, robotics, search and rescue; see Bhaskara et al., 2020; Sargent et al., 2023; Van de Merwe et al., 2024 for reviews). We view this replication of the benefits of increased transparency on automation use accuracy as a crucial contribution to the literature. The importance of establishing the robustness and generality of psychological effects is well-recognized by scholars (e.g., Pashler & Wagenmakers, 2012; Yong, 2012), and synthesizing findings across multiple studies can be used to derive more precise effect size estimates and in doing so facilitate robust evidence-based work system design (Cumming, 2012; Jones et al., 2010).

Furthermore, unlike prior UV studies (Bhaskara et al., 2021; Gegoff et al., 2024, 2025; Mercado et al., 2016; Stowers et al., 2020; Tatasciore et al., 2023; Tatasciore & Loft, 2024; but see Tatasciore et al., 2024), as well as prior studies testing automation transparency designs in other task domains (e.g., Göritzlehner et al., 2014; Guznov et al., 2020; Sadler et al., 2016), we demonstrated that, in addition to improving automation use, increased transparency benefited all other measured outcome variables (i.e., correct decision time, perceived workload, trust, and usability). Thus, the current study provides strong evidence for the potential application of the automation transparency design principle to improve human-automation teaming outcomes.

4.2. Trust calibration feedback

We predicted that trust calibration feedback would assist humans with re-evaluating their current degree of trust in the automation's capabilities and thus minimize inappropriate reliance on advice, increasing the accuracy of automation use. When trust calibration feedback was absent, participants only received feedback regarding their decision accuracy and automation accuracy on the previous trial. With trust calibration feedback, participants were also explicitly notified whether their trust was appropriate or not, and if their trust was inappropriate, were instructed to take their time and check the low/high transparency information more carefully in the future.

Contrary to our predictions, we found no benefits of trust calibration feedback on the accuracy of automation use or other outcome variables. Across all trials, correct decision times were not significantly slower for conditions in which trust calibration feedback was present ($M = 12.60$s) compared to absent ($M = 12.03$s). However, the outcomes of the trust calibration feedback questionnaire revealed that 68% of participants reported "slowing down" or "paying more attention" on future trials (missions) after receiving inappropriate trust feedback. Albeit we cannot rule out the presence of demand characteristics (Hawthorne effect; Adair, 1984) on the self-reports made by participants post-task completion. Furthermore, in line with the post-error slowing phenomenon, we predicted that decision times would be slower on the trial following incorrect compared to correct automation use, but that this effect should be more pronounced when trust calibration feedback was present compared to absent. Decision times were slower on the trial following incorrect compared to correct automation use, but participants did not slow down further when trust calibration feedback was present compared to absent. As such, beyond the post-error slowing effect, we cannot conclude that participants further slowed down following inappropriate trust calibration feedback.

A potential explanation for these null effects is that when trust calibration feedback was absent, in line with our prior UV management task studies (e.g., Gegoff et al., 2024; Tatasciore et al., 2023; Tatasciore & Loft, 2024; Tatasciore et al., 2024), feedback was still provided regarding human and automation decision accuracy. Although, when trust calibration feedback was present, it was explicitly stated whether participant trust was appropriate and *what actions to take* when trust was inappropriate, participants could still potentially

deduce their trust calibration from the information provided when trust calibration feedback was absent. We deliberately decided to employ the baseline feedback that was used in our prior studies to continue our programmatic research, but future research could further examine the benefits of trust calibration feedback in UV management compared to a baseline condition where no feedback is provided.

Overall, these findings are inconsistent with those by Okamura and Yamada (2020) who found that presenting trust calibration cues when inappropriate trust was detected (as opposed to only presenting real-time cumulative automation reliability information) increased trust calibration, as indicated by improved automation use. A key distinction between the current study and Okamura and Yamada is that in their study participants only received trust calibration cues when over-trust was detected (i.e., automation misuse), whereas the current study provided trust calibration feedback following each trial (i.e., appropriate and inappropriate trust calibration feedback), compounded by the fact that the trust calibration feedback always used the same phrases and were thus potentially easy to discount or ignore. The *adaptive* trust calibration cues used by Okamura and Yamada may have been more effective (salient) than the current study design at cueing humans to re-evaluate their trust when it was inappropriate. Future research could further examine the benefits of only presenting trust calibration cues when inappropriate trust is detected (i.e., both over/under-trust) on the accuracy of automation use.

Furthermore, in the Okamura and Yamada (2020) study when participants received trust calibration cues, they were also provided with real-time cumulative automation reliability information. However, in the current study, participants were only aware of the automation's reliability on the previous trial, rather than the cumulative reliability. Strickland et al. (2024; also see Hutchinson et al., 2023) found that human estimates of future automation reliability are disproportionality weighted by *recent* automation reliability. As such, it may be the case that in the Okamura and Yamada study, cumulative reliability information when presented together with trust calibration cues, assisted participants to further calibrate their trust to the automation's true capabilities. Future research could further examine the benefits of trust calibration cues/feedback as a function of whether cumulative automation reliability information is absent or present.

4.3. Trust calibration feedback and automation transparency

We predicted that transparency and trust calibration feedback would interact to further benefit the accuracy of automation use, given that trust calibration feedback should prompt participants to more carefully assess the highly transparent information when necessary (i.e., when their trust in automation has been historically inappropriate). This prediction was not supported. Broadly, these findings are in line with those by Tatasciore et al. (2024; also see Zhang et al., 2020) who found that case-by-case automation confidence information (i.e., automation's confidence in its advice) did not interact with transparency to further improve automation use accuracy. Although participants were cued either when trust was inappropriate (current study) or when automation confidence was low (Tatasciore et al., 2024) to further scrutinize the transparency information, across the two studies these cues did not interact with increased transparency to assist with verification of automated advice. One potential limitation of the Tatasciore et al. (2024) and current study designs is that the confidence and trust calibration cues were not designed to assist participants with determining *what* high transparency information to further scrutinize for deciding whether to rely on the advice. Future research could examine whether cueing attention to specific relevant high transparency information (i.e., projected outcome of context-specific environmental factors) when necessary (i.e., when historical trust has been inappropriate, or automation indicates low confidence in advice) can further improve the accuracy of automation use. This could also involve the presentation of other forms of local explanations (e.g., counterfactual cases) or reiteration of global explanations (Naiseh et al., 2023).

4.4. Limitations and conclusions

While the UV management task used in this study offers a broad representation of decision support in complex environments, it did not test expert operators and did not capture the same level of information complexity and multitasking demands of UV management in the field. Future research could examine the effects of transparency and trust calibration feedback using experts and higher-fidelity simulations, although achieving adequate statistical power and experimental control in such studies poses notable challenges. Although automation bias can occur in both novices and experts (Mosier et al., 1992; Skitka et al., 1999), novices tend to exhibit greater reliance (potential automation bias) than experts who typically would have greater familiarity with baseline automation reliability and context (e.g., Goddard et al., 2012). It is also the case, however, that the high experience of experts may increase automation bias due to task familiarity and associated outcome expectation (Riley, 1996). Furthermore, in field settings performance feedback is often delayed or intermittent, which could pose challenges to presenting timely (effectual) feedback or trust calibration cues. Future research could examine the effects of immediate versus delayed feedback on trust calibration.

To minimize potential fatigue effects, participants were encouraged to take breaks between training presentations and blocks of trials, and the presentation order of transparency condition (within-subjects manipulation) was counterbalanced. However, future research could also consider the effects of time-on-task fatigue (not measured here) to examine whether fatigue is associated with the use of strategies to minimize task-directed effort (Hockey, 2012), and whether fatigue increases automation reliance, although the evidence is mixed (Wohleber et al., 2019).

In conclusion, increased transparency benefitted the accuracy of automation use, decision time, perceived

workload, trust, and usability. It also led to a bias towards agreeing with advice and thus no improvement to automation correct rejection rates. Trust calibration feedback had no benefit on the accuracy of automation use and did not interact with increased transparency to further improve the accuracy of automation use.

Ethical approval

Ethics approval was received from the Human Research Ethics Office at The University of Western Australia (2019/RA/4/20/5647). Informed consent was obtained from each participant.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data used and/or analyzed during the current study are available from the corresponding authors on reasonable request.

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